Cse starter pack : everything you need to Know before 1st yr

Why Foundations of Engineering?

Because Great Engineers Don’t Just Graduate — They Prepare Early

Before all this : do this just for fun or to explore and learn , and please don’t do this as some kind off subject or course for marks or certificate

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What are foundations?

Knowing your branch is the first thing you should do. Know your curriculum. Understand your college’s style of teaching (if your college is already fixed). Try to get a glimpse of different things. Talk to a known peer who is a senior in the same field — you’ll get a better idea.

And most importantly, **don’t stress too much.**

It’s good to be first and fast, but remember to learn and go slow (not in a hurry). Try to find your current status — where you are — and **don’t give up.** Learning new things feels so different at first, and you might feel like you’re not made for it. But just **give it time.**

Concepts to Cover as Foundations for CSE:

CSE is all about computers, programs, and software. So:

- **Know about them.**
- Explore your curiosity — because the moment you hear “CSE” or see a computer, you’ll get doubts or questions.
- Ask things like:
  - How is this working?
  - How am I typing?
  - How is the computer turning on when I click a button?
  - How am I seeing pictures on the screen?
  - How are apps made?
  - How are games made?

Just explore.

How to explore?

Ask yourself. Ask your friends, peers, mentors, or guides. Use the most famous tech tool now — **AI**.

By doing this, you’ll also build the habit of using AI. But remember: **do your own work first.** Don’t entirely depend on AI. Increase your “HI” (Human Intelligence) too(that ability to think and find solutions and get to that point of solving by own)

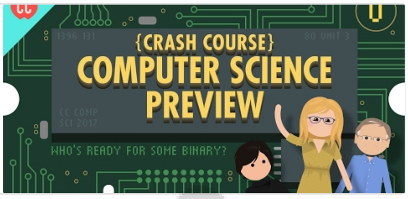
Just know what tools are present and how to use them — like ChatGPT, Perplexity, NotebookLM. Simply search on YouTube or Google how to use these tools or to get answers to your curious questions.

One such yt playlist is this

Computer Science

In 40 episodes, Carrie Anne Philbin teaches you computer science! This course is based on introductory college-level material as well as the AP Computer Scie...

https://youtube.com/playlist?list=PL8dPuuaLjXtNlUrzyH5r6jN9ullgZBpdo&si=n7IILLNnctQW9QO0



A Note About AI:

Always use AI to **save your time and effort**, but not in a way where you contribute nothing.

Know the basics, and **understand what AI is doing for you.**

Be able to **judge and modify** the output given by AI.

Some personal suggestions for AI in YT :

For Englihs or hindi language

Ansh Mehra (The Cutting Edge School)Channel: The Cutting Edge School:

https://www.youtube.com/channel/UCZknyXD-6tETm9aN8GQCi8g

For telugu language
Vamsi BhavaniChannel: Vamsi Bhavani
https://www.youtube.com/VamsiBhavani

And you can just search topic name and language you prefer and use filter options to get good results

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Here are the traditional foundation topics that we felt are good to cover:

Programming Fundamentals

- Know the **essence** of programming — try solving small, easy problems.
- Go from **flowcharts** to **pseudocode** and then to real code.
- Learn about:
  - Functions and procedures
  - Basic syntax and coding practices

Start with **C** — later, pick up other languages. Or C++ too but prefer C

Fix one language first and learn it well.

If you're into AI, ML, or Data Science — go for **Python**, but even then, I suggest learning C because it's the **birth** of many languages, tools, and software.

Choose any one... just complete it I mean learnt it for real

English or hindi

C Language Tutorial for Beginners (With Notes + Surprise) 🔥

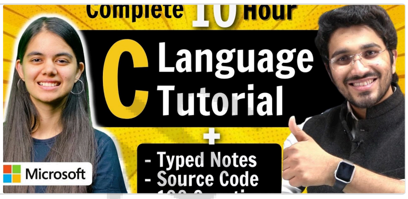
Python Udemy Course: https://goharry.in/python
Get this course at 90% Discount if you use this link

https://youtu.be/aZb0iu4uGwA?si=UtTJk8L3csZy3RYs

C Language Tutorial for Beginners (with Notes & Practice Questions)

You can join the NEW Web Development batch using the below link.
🔥Delta 3.0(Full Stack Web Development) : https://www.apnacollege.in/course/delta-batch-3

https://youtu.be/irqbmMNs2Bo?si=wk6TZrddhgSfPMVq



Telugu

C Language Full Course in Telugu | C Tutorials in Telugu | C for Beginners in Telugu

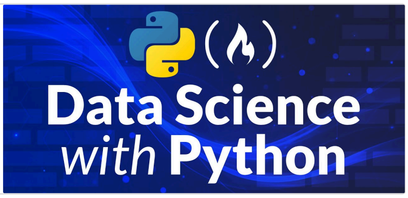
Welcome to our YouTube channel! In this educational video, we bring you a complete course on C programming language in Telugu. Whether you are a beginner or seeking to enhance your coding skills, this comprehensive course has got you covered. Dive into the

https://youtu.be/uLfCFa1Vaj0?si=ypaYyqNRO5u1RihS

Learn Python for Data Science – Full Course for Beginners

Get started with data science using Python! This course covers essential tools like Pandas and NumPy, plus data visualization, cleaning, and machine learning techniques. Perfect for beginners, you'll gain the skills to analyze and interpret data effectively.

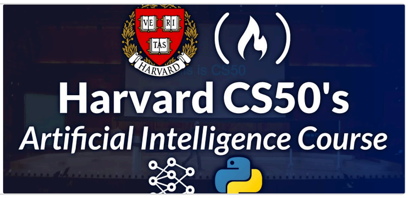
https://youtu.be/CMEWVn1uZpQ?si=_1CxxpasFbYLmQe4



Harvard CS50's Artificial Intelligence with Python – Full University Course

This course from Harvard University explores the concepts and algorithms at the foundation of modern artificial intelligence, diving into the ideas that give rise to technologies like large language models, game-playing engines, handwriting recognition,

https://youtu.be/5NgNicANyqM?si=o3JCF2_NrG4Q4LRd



Python Language Full Course (2025-26)

https://youtube.com/playlist?list=PLGjplNEQ1it8-0CmoljS5yeV-GlKSUEt0&si=8l_QKU9HWnvUz_xp

Side Qn:

> When and which programming language was discovered first? How did it evolve?

Remember: Once you master **one** language, you can learn or shift to other languages **very easily** (trust me on this).

Know Your Computer

- Parts, connections — just explore it.
- Start from: What is a computer?
- What are its parts?

- How are they connected and how do they work together?

Computer Functionalities

- Memory management and data utilization
- Execution flow and control mechanisms

100+ Computer Science Concepts Explained

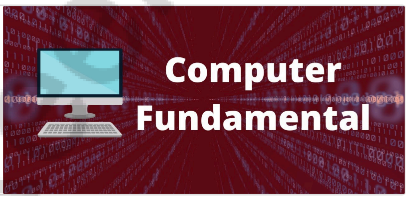
Learn the fundamentals of Computer Science with a quick breakdown of jargon that every software engineer should know. Over 100 technical concepts from the CS curriculum are explained to provide a foundation for programmers.

https://youtu.be/-uleG_Vecis?si=EBILgWmUAAwZGr7T

Computer Fundamentals - Basics for Beginners

A computer is an electronic machine that accepts data, stores and processes data into information. The computer is able to work because there are instructions in its memory directing it.

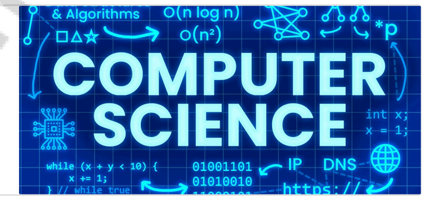
https://youtu.be/eEo_aacpwCw?si=ypBOf4EJC6joE5sR



COMPUTER SCIENCE explained in 17 Minutes

Learn more about Computer Science, Math, and AI with Brilliant! First 30 Days are free + 20% off an annual subscription when you use our link: https://brilliant.org/WackyScience/

https://youtu.be/CxGSnA-RTsA?si=m5QOl6lEH5w43Or5



Introduction to Programming and Computer Science - Full Course

In this course, you will learn basics of computer programming and computer science. The concepts you learn apply to any and all programming languages and will be a good base onto which you can build your skills.

https://youtu.be/zOjov-2OZ0E?si=a9jeol8SXby_UdM0

Computer Science

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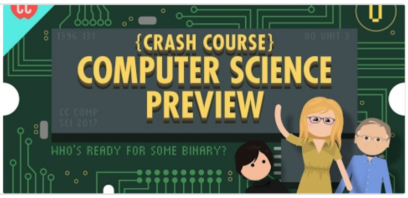


Operating Systems Course for Beginners

Learn fundamental and advanced operating system concepts in 25 hours. This course will give you a comprehensive understanding of how operating systems function and manage resources.

https://youtu.be/yK1uBHPdp30?si=LyW9lwudhkV339GE

Learn About What Actual CSE Graduates / Software Developers Do

Remember: **You are just starting.**

So it's okay to know **only something** at this point.

Go through concepts. Watch videos or lectures. You'll learn them in more depth during your course.

And if you **get it** and want to go deeper — **congratulations**, you can go.

If you’re getting many questions like:

“Why this?”, “How that?”, “What is this?” — it’s normal.

You might feel like “everything is IDK” or “I’ll only know later” — that’s okay.

You’re free to explore and dig deeper.

But ask yourself: “**Will this help me at present if I invest my time in it?**”

Everything will come at its right time.

So just trust and believe — do what’s required **now** and move on.

(Learning to accept some things “as is.”)

Applications of Computer Fundamentals

- **Software Development**: Version control, IDEs, debugging techniques
- **Network Administration**: Basic networking protocols, TCP/IP

Software Essentials

- Types of software: Application vs. System Software
- Operating systems fundamentals

What do software engineers/developers do? Software engineers ఏం చేస్తారు? Vamsi Bhavani

In this video, we will try to understand what software engineers or software developers do? We will also try to understand their work and see how it differs between software engineers in service based companies and software engineers in product based campanies. This video will be very insightful for

https://youtu.be/uGQWpcO30eI?si=TJokLbj37s_d9GuK

What Do Software Engineers ACTUALLY Do?

In this video, I will talk about what software engineers actually do all day.

Software engineering is much more than just sitting behind a computer 8 hours a day and

https://youtu.be/ilxZrYzJJ7I?si=N8I0Kj6aiZejdQeK

What Do Software Engineers Actually Do? (It’s Not What You Think)

In this video, I will discuss about what exactly you will do after becoming a software engineer in a big company. This video will throw light on what the role of a software engineer is!

https://youtu.be/HDXpGgieFDg?si=ImAc2wM5afidexTt

Software Engineering (SE / SEPM)

Hii.. 👉SE Important Topics Videos 👉 SE Important Questions LIKE, SHARE & SUBSCRIBE 🙏 Thank You. 😊 #softwareengineering #softwareenginerringtutorial #software...

https://youtube.com/playlist?list=PLQ-nEJNYIEV29CBLzIDxcogm6CEZjVad2&si=4jgoj0-sjcstt9nA

Problem Solving: DSA

Most things are about **problem-solving.**

And in CSE, to get that mindset, we do **DSA**.

Do it with some fun.

Start from 0 — like "add two numbers."

Go step by step at your **own pace**.

Please **don't just mug up** the concepts.

Don't just complete lectures or sheets.

Don't copy-paste answers.

And please don't use AI to write code —

You **can** use it to learn different approaches,

but **do it yourself first.**

Don't do this just for placements.

(Yes, DSA plays a big role in placements.)

But do it to **improve your core skill of problem-solving**.

You have a **lot of time** — so start from 0 and grow slowly.

Before starting DSA, answer these:

- What is data? What is its importance?
- What is information?
- What is problem-solving?
- What are structures?
- What is data in CSE?
- What are algorithms?
- What is DSA?

Now Start Learning DSA:

Pro tip:

Go topic by topic. Solve as many questions as possible.

Try to find **patterns** yourself.

After some decent progress (5–6 topics), explore **CodeChef**, then later move to **LeetCode**.

Fix one question sheet — no matter what.

It's all about: **Can you solve problems?**

Data Structures and Algorithms

- Basic sorting and searching algorithms
- Time and space complexity introduction

Choose one and complete it.. Don't try to all

Strivers A2Z-DSA Course | DSA Playlist | Placements

This playlist is not in C++ or Java or Python, it covers DSA and we write pseudocode, only one video is in C++, for the basics, but apart from that we cover ...

https://youtube.com/playlist?list=PLgUwDviBIf0oF6QL8m22w1hIDC1vJ_BHz&si=LupLw2eT7pRZIWKp

Complete C++ Placement DSA Course

https://youtube.com/playlist?list=PLDzeHZWIZsTryvtXdMr6rPh4IDexB5NIA&si=1Rozwe0FtZ2CRJer

Complete C++ DSA Course | Data Structures & Algorithms Playlist

#1 DSA Series for Placement Preparation in Making. Neither less nor more : Study only what's needed & Save Time!

https://youtube.com/playlist?list=PLfqMhTWNBTe137I_EPQd34TsgV6IO55pt&si=DHERwIgIETd6AfyJ

Love Hacking Stuff from Movies?

We all do!

Why not explore it a bit?

Use Google, YouTube, and AI tools. Learn about:

- Hacking
- Cybersecurity
- Good practices in digital safety
- How to keep your digital data & devices safe
- How to learn ethical hacking and cybersecurity

7 Cybersecurity Tips NOBODY Tells You (but are EASY to do)

The exact same security tips get repeated all the time. Of COURSE you want strong passwords and 2FA, but what are some easy steps you can take (less than 5 min!) that can drastically improve your online security? If you haven't removed your personal data

https://youtu.be/Uy60wy20ADE?si=eYCBClIWHg4qmD1P

11 Internet Safety Tips for Your Online Security

Watch this video to learn how to protect yourself online. Follow these rules to not get a victim of phishing, scams, ransomware and cyber threats.

https://youtu.be/aO858HyFbKI?si=WxKY08mNQZ8MdcIF

Fundamentals of Cybersecurity

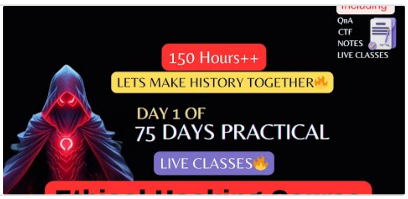
- Web Application Security Basics
- Security Awareness and Best Practices

https://youtu.be/K-WLfzXmX3U?si=T3AgCce9FvmwQ3Af

75 days Advance Ethical hacking full course with notes in hindi - 2025 Edition

🔥 Master Ethical Hacking in 75 Days! 🔥 Welcome to the ultimate free Ethical Hacking course—a 75-day intensive training program covering Windows hacking, Linu...

https://youtube.com/playlist?list=PLGjHK4PZVxhRAYh6o4iTZ94SN3W4t9aaf&si=zCMEosFo3f8XXMoD



With All the AI Buzz

- Learn about different AI tools
- Learn **basic prompt engineering**
- Learn basics of AI
- Learn how to **stay in the loop** with AI updates

Just find some weekly updates use Google and use perplexity AI to get that

And just search in YouTube about this

Soon we will share more on this seperately

ALL THE BEST